

SHOCK

Headings

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3. THIRD STAGE / IRREVERSIBLE STAGE (REFRACTORY)

- **Shock** : Refers to depression or suppression of body functions produced by any disorder.
- **Circulatory Shock** : Refers to shock developed by **inadequate blood flow throughout the body** (*a life-threatening condition*).

MANIFESTATION OF CIRCULATORY SHOCK

Insufficient blood flow to tissues (particularly brain)



Due to severe reduction in Cardiac Output (CO)

- a. ↓ **Cardiac Output (CO)** → Arterial Blood Pressure (BP) drops down.
- b. **Low BP** → Produces **reflex tachycardia** and **reflex vasoconstriction**.
- c. **Tachycardia** → ↓ Decreases diastolic period:

↓ Diastolic Filling → ↓ Stroke Volume (SV) & Systolic BP → ↓ Pulse Pressure (< 20 mmHg)
- d. ↓ **Velocity of blood flow** → Causes **stagnant hypoxia**.
- e. **Skin becomes pale and cold** due to intense vasoconstriction.
- f. **Cyanosis develops** with worsening hypoxia (*especially visible on ear lobes and fingertips*).
- g. ↓ **GFR and urinary output** due to profound fall in renal perfusion pressure.
- h. **Metabolic activities of myocardium accelerate** due to ↓ blood flow and ↑ heart rate:

Anaerobic glycolysis → Large amount of lactic acid formed → **Metabolic Acidosis**

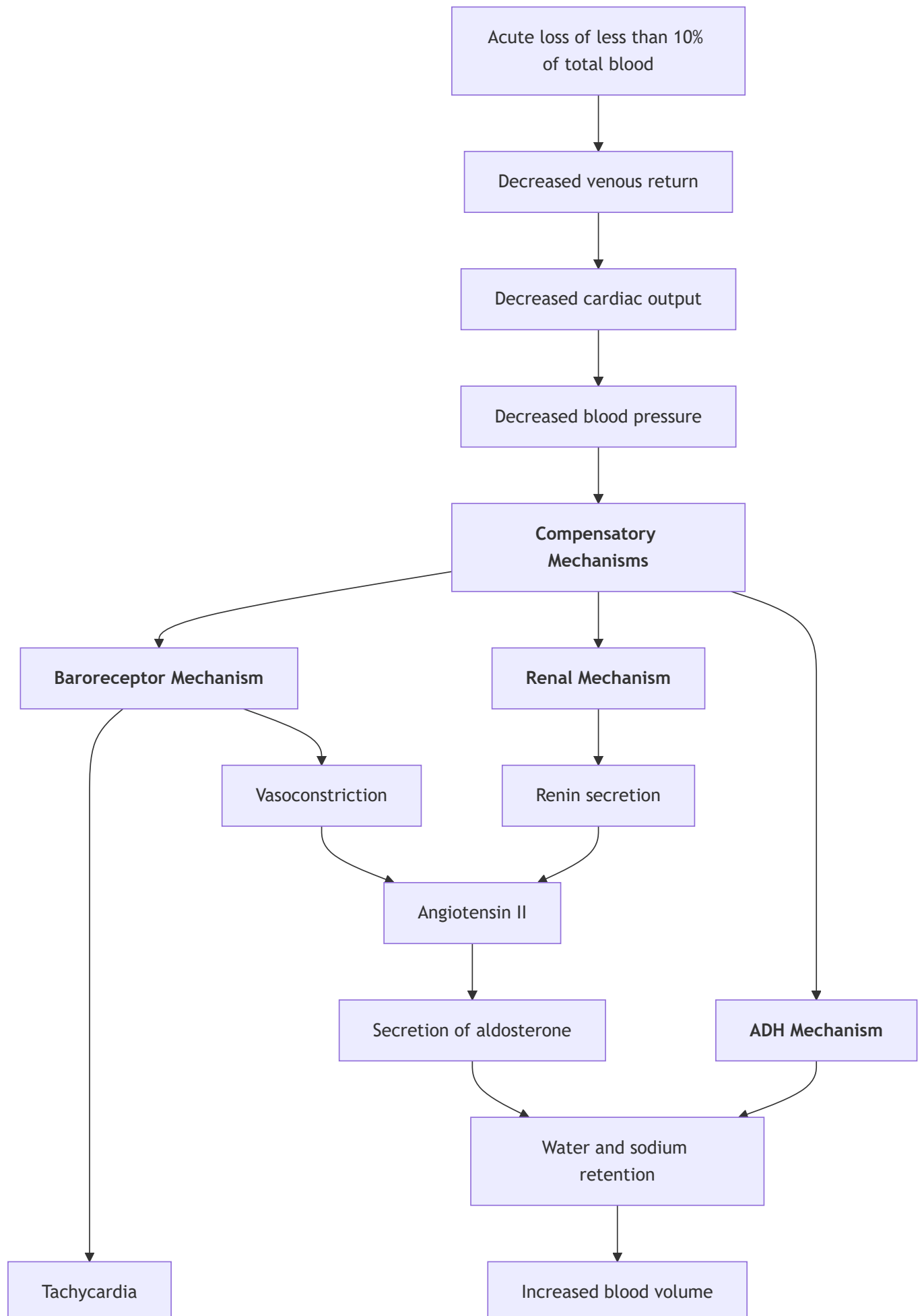
STAGES OF CIRCULATORY SHOCK

Circulatory shock occurs in **3 distinct stages**:

- i. **1st Stage / Compensated Stage** (*Non-progressive*)
- ii. **2nd Stage / Progressive Stage** (*Decompensated*)
- iii. **3rd Stage / Irreversible Stage** (*Refractory*)

1. FIRST STAGE / COMPENSATED STAGE (NON-PROGRESSIVE)

- **Features** : Regulated by **negative (–ve) feedback control mechanisms**; corrects blood loss of $< 10\%$ of total blood volume.
- **Compensatory Mechanisms Involved** :—
 - a. **Baroreceptor Mechanism** : Initiates strong sympathetic stimulation $\longrightarrow$ Tachycardia, vasoconstriction $\longrightarrow \uparrow$ BP.
 - b. **Renin-Angiotensin-Aldosterone (Renal) Mechanism** : Restores blood volume through vasoconstriction and Na^+ / water retention.
 - c. **Antidiuretic Hormone (ADH) Mechanism** : Potent vasoconstriction and free water retention.



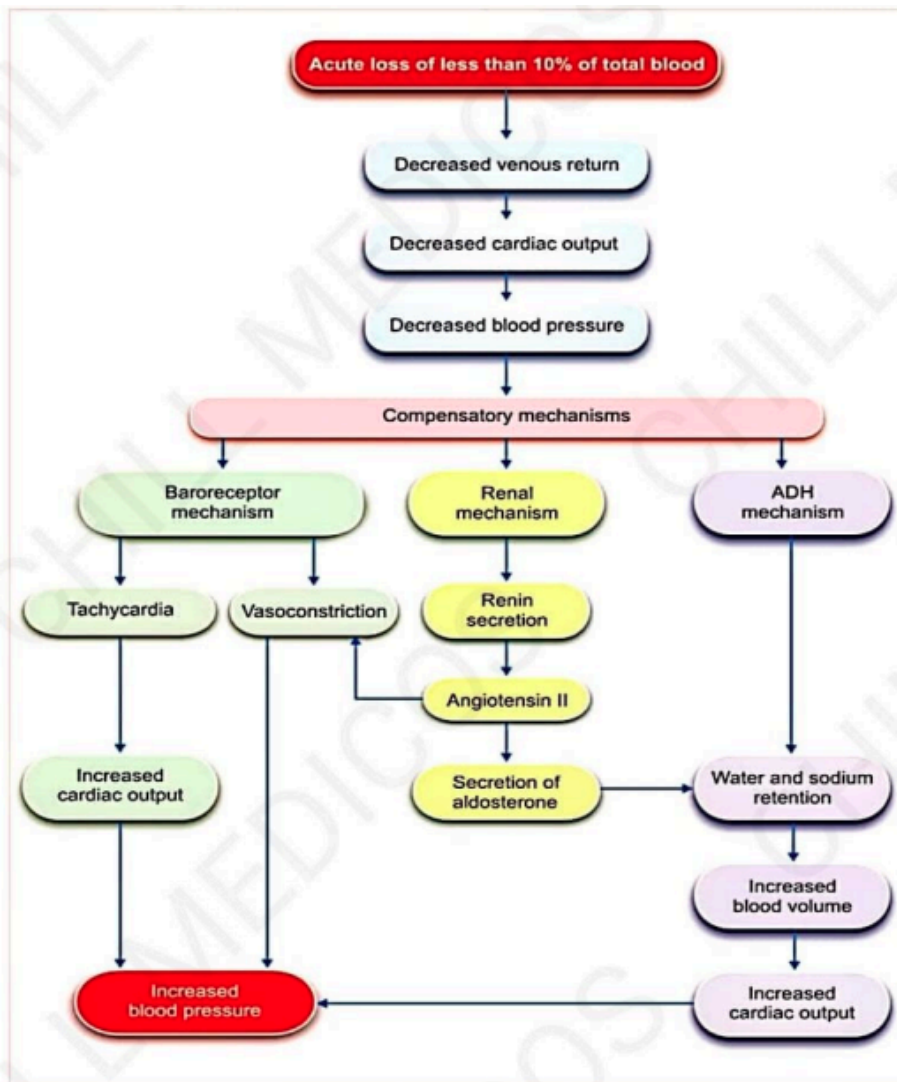
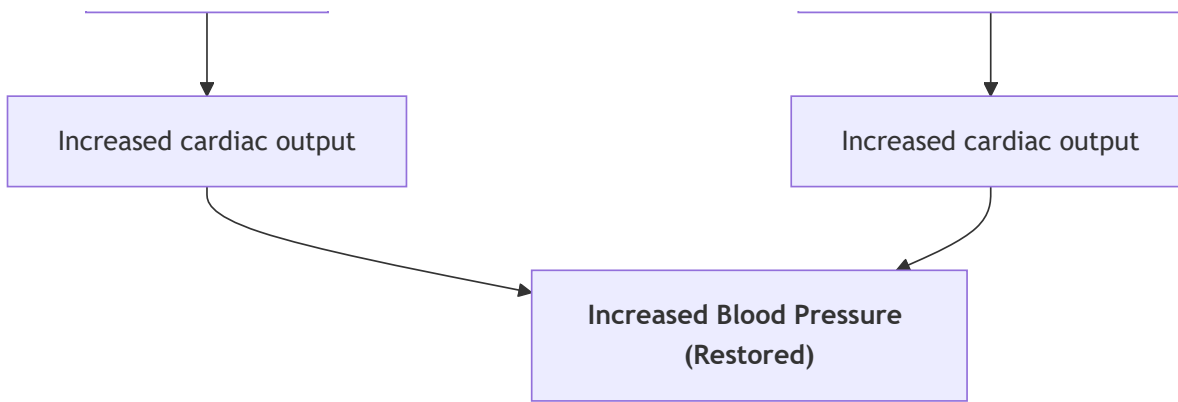


FIGURE 116.1: Compensated stage of circulatory shock. ADH = Antidiuretic hormone.

2. SECOND STAGE / PROGRESSIVE STAGE (DECOMPENSATED)

- **Features** : Occurs with 10% – 15% **acute loss of total blood volume**.
- A **positive (+ve) feedback vicious cycle** develops, making normal regulatory mechanisms inadequate to compensate.
- Bacterial toxins (**Endotoxins**) worsen myocardial depression.

Acute loss of 10% to 15% of
total blood



Severe reduction in venous
return



Severe reduction in
cardiac output



Severe decrease in blood
pressure



Suppression of vasomotor
center



Severe vasodilatation



Further decrease in blood
pressure

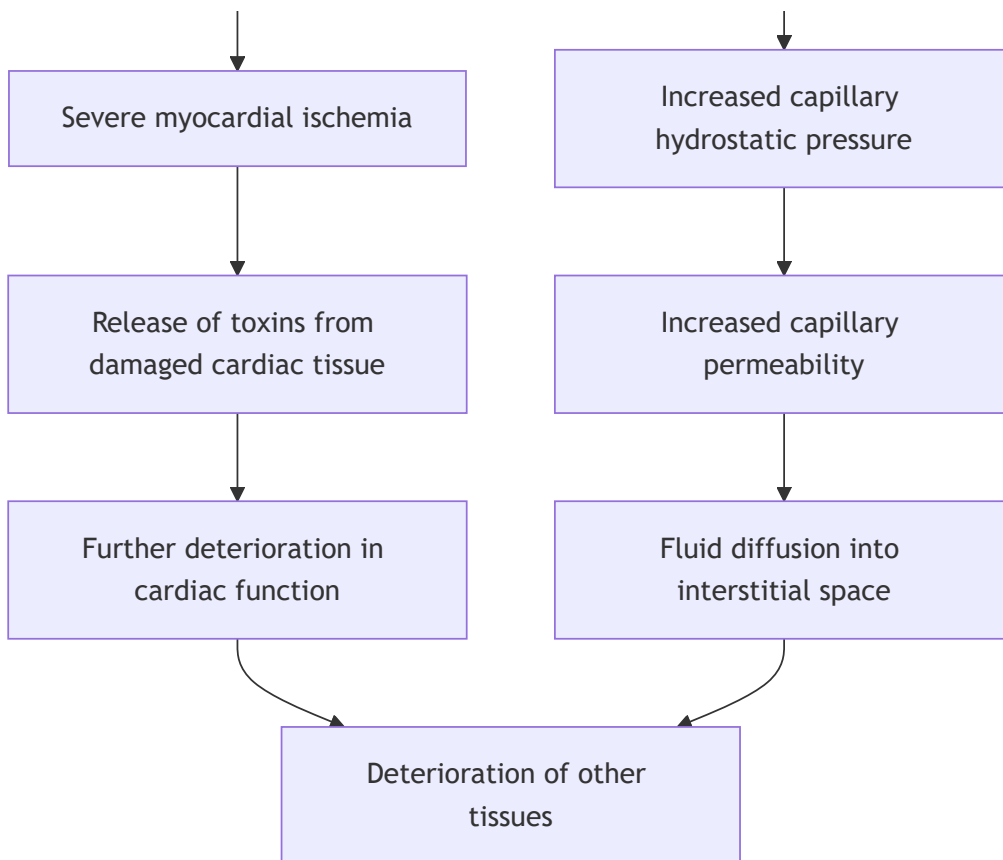


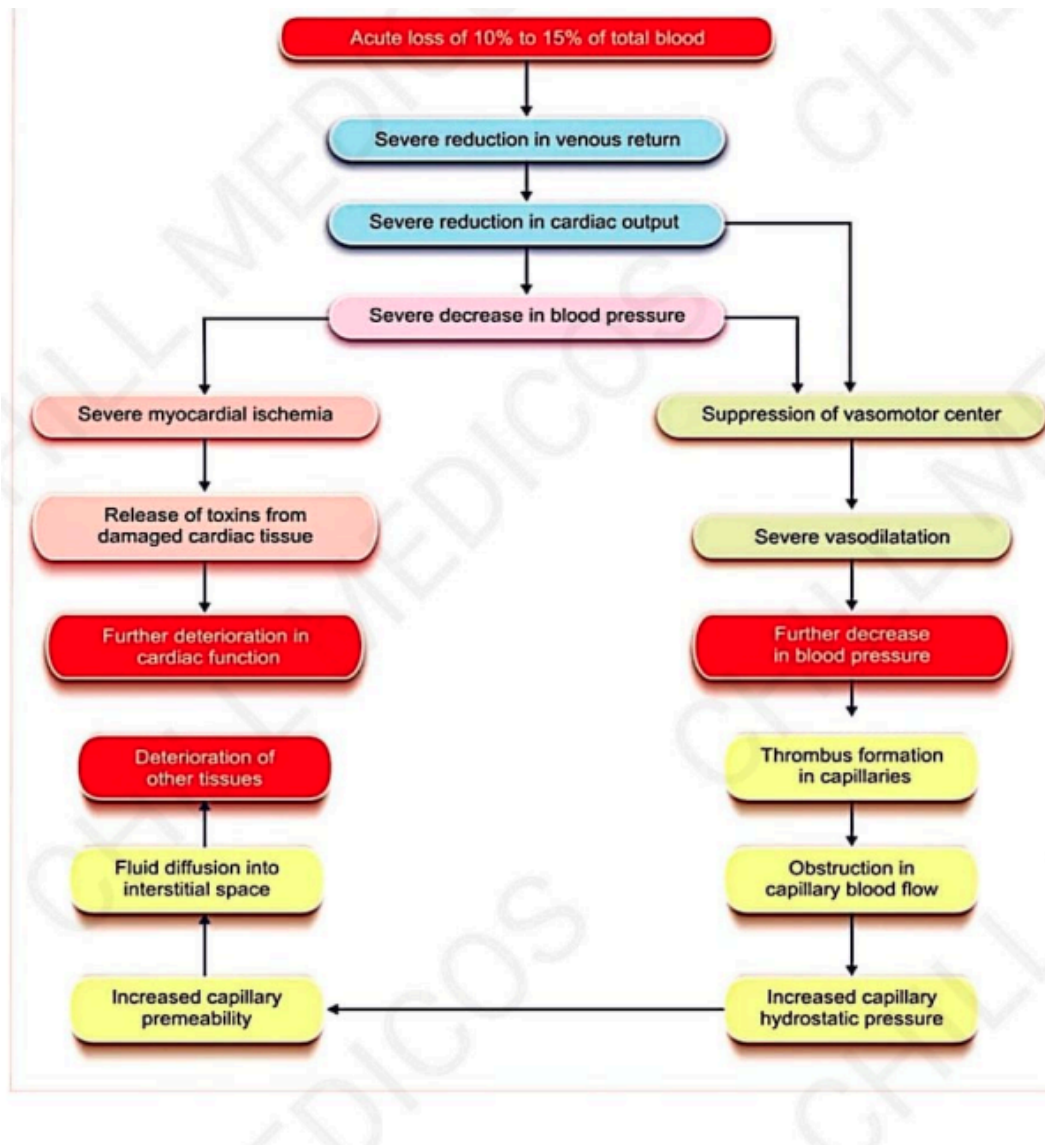
Thrombus formation in
capillaries



Obstruction in capillary
blood flow

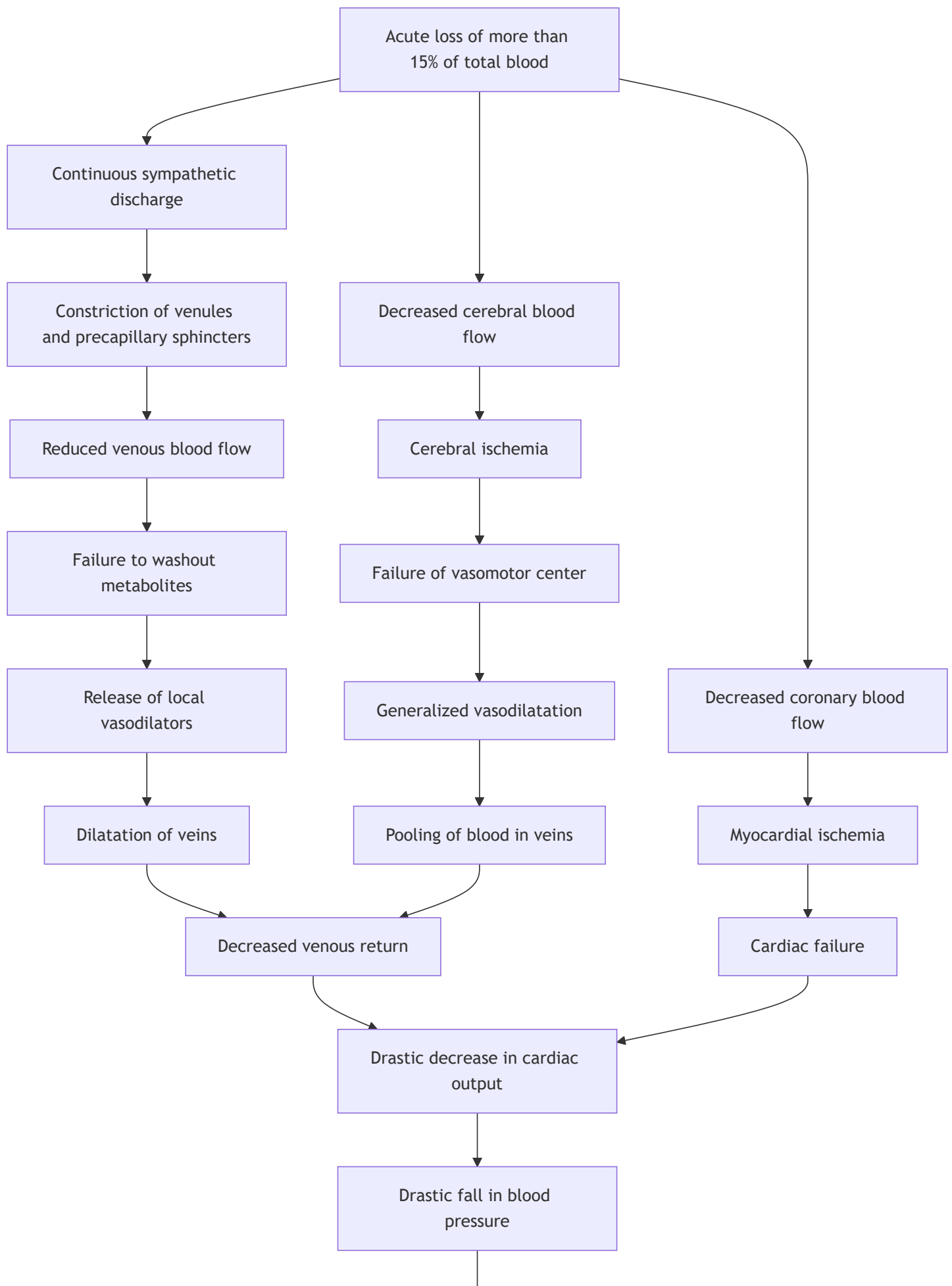






3. THIRD STAGE / IRREVERSIBLE STAGE (REFRACTORY)

- **Features** : Occurs with **acute loss of $> 15\%$ of total blood volume**.
- Also called the **Refractory Stage**.
- Leads inevitably to **death** due to complete failure of vital organ perfusion, severe cerebral ischemia, and myocardial collapse.



DEATH

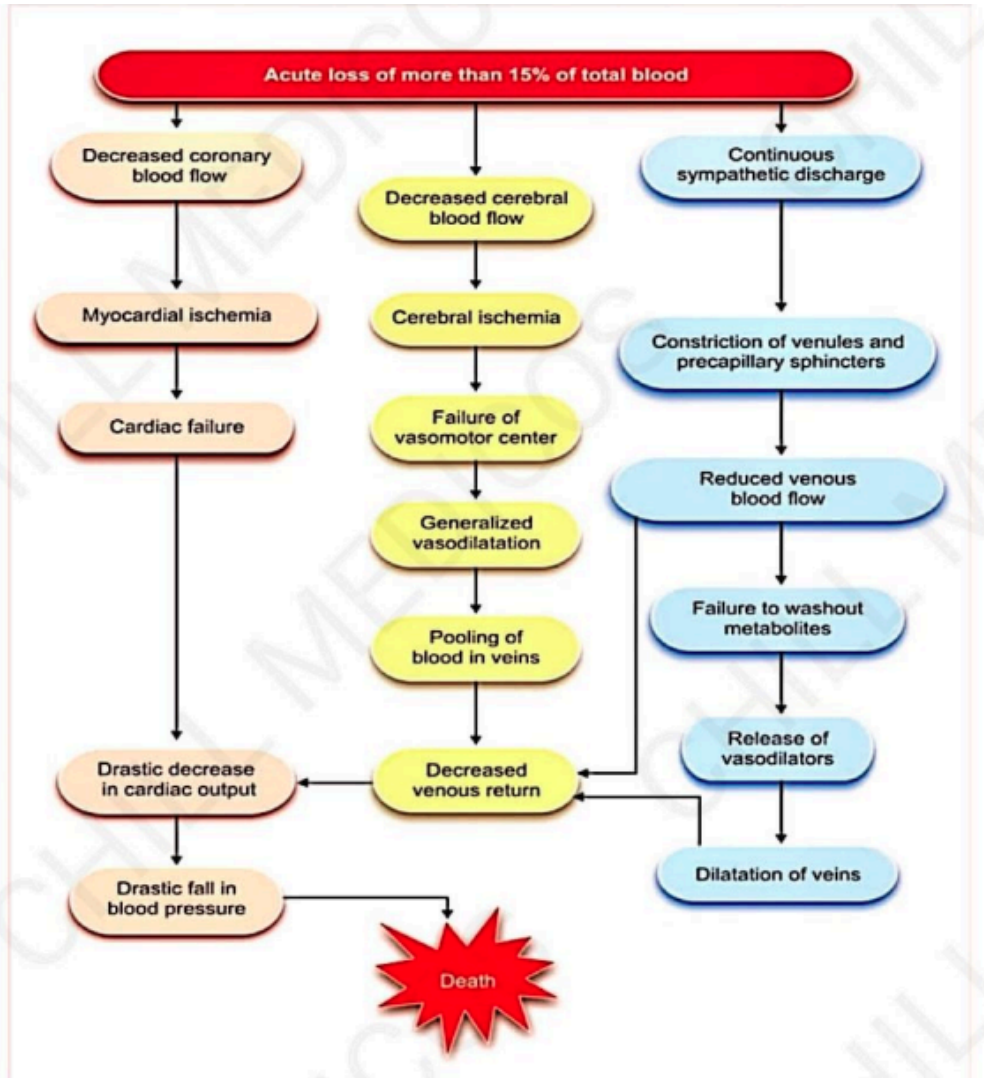


FIGURE 116.3: Irreversible stage of circulatory shock